Beginner Python Practice Tasks

Two console programming exercises using the topics covered so far

Topics used

Input and output · variables · type conversion · arithmetic · conditions · loops · strings · lists · dictionaries

Task 1

Stationery Shop Billing System

Build a small stationery shop billing program. Write it from top to bottom in one file. Use no functions, classes, or file saving.

Budget rule

Set a fixed customer budget of ₹3,000 near the beginning of the program. If the final bill is more than ₹3,000, display Can't pay: bill is over the ₹3,000 budget and stop checkout before payment.

Starting items

ID	Stationery item	Price	Available
1	Pencil	₹10	Yes
2	Eraser	₹5	Yes
3	Notebook	₹50	Yes
4	Pen	₹20	Yes
5	Geometry Box	₹120	Yes

Store these items in a list of dictionaries. Each dictionary should contain an ID, name, price, and availability.

Part 1: Show the items

Use a loop to display each item's ID, name, price, and availability. Use an if statement to show Available or Not available.

Part 2: Take one customer's order

1. Ask for the customer's name.
2. Ask for a stationery item ID and quantity. Let the customer add multiple items; entering 0 for the ID finishes ordering.
3. Search for the ID by looping through the items. If it is invalid or unavailable, show a message and ask again.
4. Do not accept a quantity of zero or less. Store each ordered item's name, price, and quantity in an order list.
5. After each item, show its line total (price × quantity) and the running subtotal. Do not allow checkout for an empty order.

Task 1: Billing and checkout

Part 3: Calculate the bill

Ask whether the customer wants to pay by cash or online. Online payment gets a 5% discount on the subtotal; cash gets no discount. Calculate 5% tax on the amount after any online discount. The final bill is subtotal - online discount + tax.

Show the subtotal, online discount (or zero for cash), amount after discount, tax, and final bill with two decimal places. Compare the final bill with the ₹3,000 budget. If it is over budget, show the budget message and stop checkout; otherwise continue to payment.

Example: subtotal ₹600, online payment

Subtotal	Online discount	After discount	Tax	Final bill
₹600.00	₹30.00	₹570.00	₹28.50	₹598.50

Part 4: Payment and receipt

Print a receipt with the stationery shop name, customer name, each item, quantity, line total, subtotal, online discount, tax, and final bill.

Part 5: Simple summary

- Number of different items in the order and total quantity
- Total bill amount and whether it was within budget
- Most expensive item in the shop, found using a loop and comparison

Practice rules

- Use loops and conditions to search and calculate; do not calculate the bill manually.
- Use clear variable names and helpful messages.
- Handle wrong IDs, unavailable items, invalid quantities, and insufficient payment without crashing.

Task 2

Print a Descending Star Pattern

Write a separate Python program that prints this pattern using for loops:

```
* * * * *
* * * *
* * *
* *
*
```

Requirements

- Use an outer for loop to control the rows and a nested for loop to print stars.
- Print five stars on the first row, reducing the count by one on each following row.
- Print each row on its own line. Do not use functions, classes, or pattern libraries.